

JIEMING ZHANG

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EDUCATION

University of Virginia

Charlottesville, VA

B.S. Computer Engineering & B.S. Electrical Engineering, **GPA : 3.9**

Expected Graduation: May 2029

Relevant Coursework: Computer Systems Organization, Electric Circuits, Digital Logic Design, CPU Architecture, Ordinary Differential Equations, Multivariable Calculus, Data Structures & C++

TECHNICAL SKILLS

Languages: C++, Python, VHDL, Java

Hardware & EDA: Custom PCB Design, Power Management (3.7V LiPo to 3.3V LDO), KiCad, Altium, Quartus, ModelSim, ESP32, BNO055 IMU, MicroSD SPI, Oscilloscopes, Multimeters

Protocols & Tools: I2C, SPI, UART, CAN Bus, Fusion 360, Git, Matplotlib, SciPy

Concepts: Low-Level Firmware, Digital Logic, CPU Architecture, Signal Processing, Sensor Fusion

TECHNICAL PROJECTS

IoT Golf Swing Analyzer & Visualizer Prototype

Summer 2026

Embedded Hardware, Firmware & Python Analytics · Personal Project

- Designed and routed a custom ESP32 PCB with 3.7V LiPo-to-3.3V power management and interfaces for the IMU and MicroSD logger.
- Built non-blocking C++ firmware to log 100 Hz BNO055 IMU quaternion and accelerometer data over I2C to SPI MicroSD storage.
- Created a Python telemetry engine using Matplotlib and SciPy to process raw quaternions, correct gravitational drift, estimate clubhead velocity, and render 3D swing trajectories.

CPU Datapath & Controller Design

Spring 2026

UVA ECE Coursework

- Architected a structural VHDL CPU datapath with ALU functions, register banks, and control sequencers.
- Debugged timing and control-state faults in ModelSim using testbench reset pulses and waveform traces; synthesized the design in Intel Quartus.

Automated Physical Sink Shutoff System

Fall 2025

Introductory Engineering Design Project · UVA

- Engineered a retrofit automated shutoff mechanism for standard sink handles without internal plumbing access.
- Integrated an ultrasonic distance sensor to detect user presence and trigger automated shutoff sequences.
- Designed and calibrated a servo-driven mechanical arm to actuate the external water valve handle.

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE)

Fall 2025 - Present

Data Acquisition (DAQ) Subteam Member | University of Virginia

- Design schematics and multi-layer PCBs in Altium and KiCad for sensor integration and data logging on an all-electric FSAE racecar.
- Build wiring harnesses and model custom DAQ mounting enclosures in Fusion 360 for noise-resistant vehicle telemetry.
- Integrate sensor and chassis-hardware components for reliable vehicle installation.

HACKATHON EXPERIENCE

VTHacks 14

Sep 2026

2-person team · Embedded Systems & Wireless Sensing

- Built an ESP32 occupancy-sensing prototype that scanned 2.4 GHz Wi-Fi and Bluetooth traffic for packet counts and RSSI.
- Developed C++ firmware to compute a 0-100 density score from signal activity and fluctuation, calibrated against an empty-room baseline stored in nonvolatile memory.
- Produced device measurements for a partner-developed Supabase and React/Node.js dashboard that visualized occupancy trends.